

Too Soon to Dismiss? A Reanalysis of Forster and Neugebauer (2024) Highlighting the Role of Effect Heterogeneity in Evaluating Factorial Surveys

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Abstract

This replication study critically examines the findings and conclusions of Forster and Neugebauer (2024), who question the external validity of factorial surveys in capturing real-world behavior. Using the publicly available data from the original study, we reassess the effects of educational signals and ethnic origin. Our analyses show that the original results do not withstand all robustness checks and that theoretically derived effect heterogeneities across occupational fields provide more nuanced explanations. These findings challenge the general claim that factorial surveys are unsuitable for studying hiring processes and instead underscore the importance of context-sensitive modeling. We argue that factorial surveys remain a valuable methodological tool, particularly when heterogeneity is explicitly addressed in both design and analysis.

Introduction

In their recent validation study, Forster and Neugebauer (2024) conduct two highly parallelized experimental studies to investigate whether results from factorial surveys (FS) can be transferred to understand real-world decisions (see also Petzold and Wolbring, 2019; Pager and Quillian, 2005). They focus on two differently sensitive factors influencing hiring chances into an apprenticeship position: educational signals and ethnic origin. By choosing personnel selection as their research context, they situate their study within a field where both field experiments (FE) and FS are frequently applied (e.g. Di Stasio and McDonald, 2025; Schimke, 2023; Piopiunik *et al.*, 2020).

Earlier this year, Pickett's (2025) comments and a corresponding response to the critique it raised were published (Forster and Neugebauer, 2025). While Pickett's discussion centers on (necessary) design differences between the FE and the FS, most of which are addressed in Forster and Neugebauer's (2025) reply, the present replication builds on additional analyses of the publicly available data (<https://osf.io/x2tcp/>). These analyses are based on considerations from labour queue theory Thurow (1975) and statistical discrimination (Arrow, 1973; Aigner and Cain, 1977; Phelps, 1972).

In their study, Forster and Neugebauer (2024) report that in the FS, higher education dropout is associated with a significantly lower probability of being invited to job interviews than having completed only the Abitur. In contrast, the FE reveals no significant difference between the two groups. A direct comparison shows that the non-completion estimates differ significantly across the two designs. We argue (1) that this central finding does not withstand all of the extensive robustness checks reported in Forster and Neugebauer (2024). (2) Drawing on labour queue theory, we test a hypothesis regarding the relative value of educational signals and reanalyze the data from Forster and Neugebauer (2024) for effect heterogeneity across the occupational fields of the sample. (3) We conduct the same analyses for the discrimination findings and, finally, discuss the results in light of the study's central conclusion: "FSs are not suited for studying hiring behavior" (Forster and Neugebauer, 2024: 901).

Reinterpreting the robustness check on selection bias

In their supplementary material, Forster and Neugebauer (2024) present a series of valuable sensitivity analyses to explore potential explanations for the divergent findings between their FE and FS. An important concern is potential sample selection bias: "we restrict the sample of the FE to those respondents who also participated in the FS so that we look at exactly the same respondents in both experiments" (Forster and Neugebauer, 2024: S23). The result is an insignificant but in this case negative point estimate in the FE (-0.02). The authors conclude that the closer correspondence between FE and FS (-0.05) results is not enough to assume that the different results are driven by selection bias: "Therefore, this analysis does not change the conclusions we draw in the main text" (Forster and Neugebauer, 2024: S23f.).

However, unlike the main analyses (cf. Table S7), this robustness check omits a formal statistical test of the difference between the two experiments, leaving unclear why the observed divergence does not lead to a revision or at least a restriction of the study's conclusions. Conducting this test yields a difference of -0.03 with a p-value of 0.473, well below conventional thresholds of statistical significance. While the direction of the estimate aligns with the main result and the reduced sample size implies expected limitations in statistical power, a 32% reduction of the estimate at least constitutes a noteworthy restriction on the conclusions drawn.

Additional analyses on effect heterogeneity for the educational signal

The positional model of education highlights that the labour market value of educational credentials is not fixed, but depends on the relative position of a jobseeker within the broader pool of applicants (Thurow, 1975). In this job competition model, employers form a labour queue in which they rank applicants based on perceived trainability, using education as a key

sorting signal. Employers prefer candidates positioned at the front of this labour queue, which should translate into higher job interview invitation probabilities.

Within this framework, the value of a specific educational pathway, such as a higher education dropout, is inherently context-dependent. When the applicant pool contains many individuals with the same degrees, an unfinished higher education may not signal partial academic achievement, but rather failure or instability, especially when compared to applicants with a completed upper secondary qualification (e.g., Abitur) and no dropout history. The finding by Forster and Neugebauer (2024) that higher education non-completion is associated with lower hiring chances than an upper secondary school degree alone (in the FS) may thus reflect a relative disadvantage in the labour queue, rather than an absolute devaluation of their additional incomplete higher education. Building on this idea, our replication examines whether the signalling penalty for higher education dropout varies across occupational fields in both experiments, depending on the typical educational composition of applicants and the credential structure within the applicant pool. If the positional logic holds, we would expect the dropout penalty to be stronger in fields where higher education is the norm and weaker or absent in fields where an Abitur still represents a competitive advantage.

Forster and Neugebauer (2024: 893f.) seem to be aware of the problem that might arise from this considerations: „To create a comparable pool of competitors in the FS, we constructed additional applicants with an intermediate secondary school leaving certificate.” FS, however, rely on the orthogonality and balance of experimental factors and their respective levels (Auspurg and Hinz, 2015). An alignment between the applicant pools in the FS and the FE is therefore only plausible if one assumes an equal distribution of applications across the three implemented educational categories: intermediate secondary education, higher education entrance qualification (Abitur), and Abitur plus higher education dropout. Based on available data from official statistics, this assumption appears at least questionable. Table 1 presents weighted averages for the educational attainments of new apprentices in the 21 selected occupations separately and aggregated by occupational field as presented in Forster and Neugebauer (2024). In the electronic occupations, for instance, only about 43% of new hired apprentices hold a higher education entrance qualification, whereas in the fields of laboratory and media, roughly two out of three apprentices do. If the recruited apprentices in a given field tend to have relatively high school-leaving qualifications, it is reasonable to assume that the applicant pool also skews towards higher educational backgrounds.

Table 1. Additional information on the selected occupations of Forster & Neugebauer (2024).

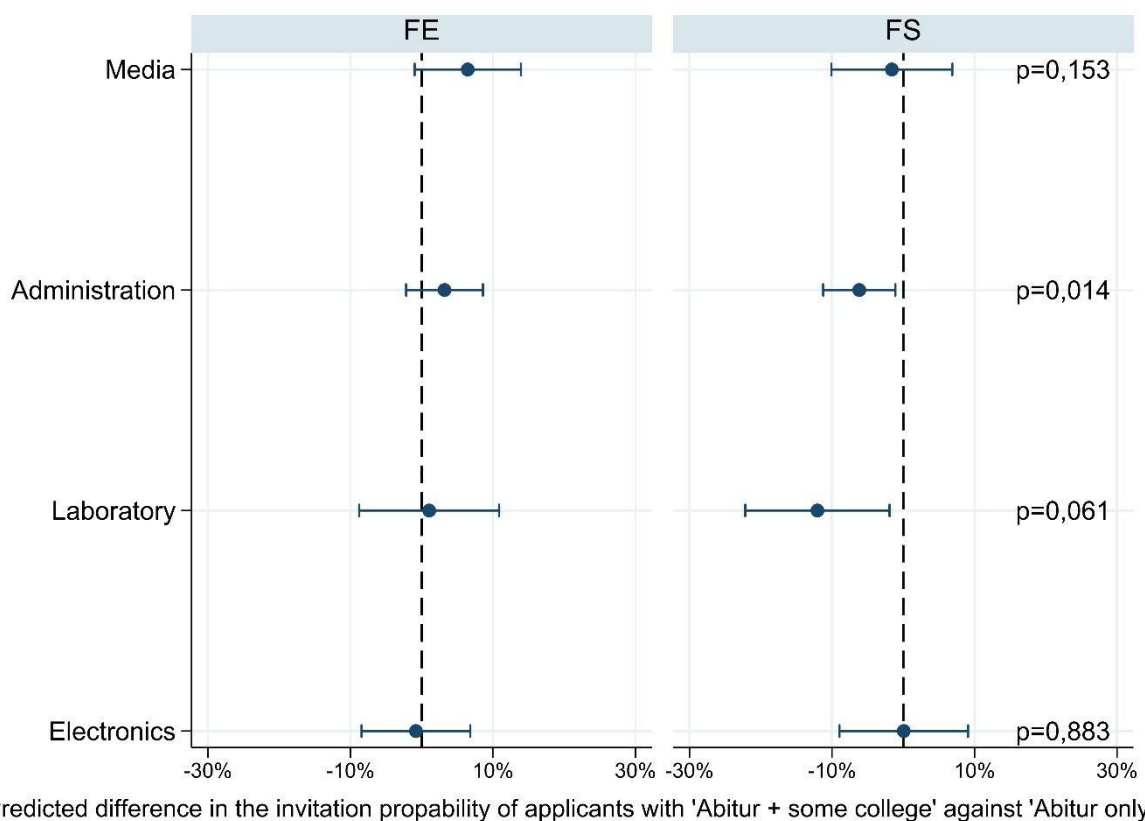
Apprenticeship occupation / Occupational field	New apprentices (2023)	New apprentices with HEEQ (2023)	New apprentices with foreign origin (2023)
Electronics technician for automation technology	2,352	44%	6%
Microtechnologist	174	53%	9%
Electronics technician for information & systems technology	186	60%	5%
Aircraft electronics technician	120	40%	5%
IT systems electronics technician	1,383	37%	7%
Mechatronic and electronic occupations	4,215	43%	6%
Biology laboratory technician	489	79%	2%
Dairy laboratory technician	150	36%	2%
Chemical laboratory technician	1,650	68%	5%
Pharmaceutical technician	273	55%	7%
Physics laboratory technician	114	63%	5%
Materials tester	270	46%	6%
Laboratory occupations	2,946	65%	4%
Administrative assistant	6,780	52%	3%
Specialist for labor market services	705	52%	6%
Social insurance clerk	2,442	60%	4%
Management assistant in healthcare	2,241	60%	4%
Specialist for media and information services	570	70%	2%
Administration occupations	12,738	56%	3%
Management assistant for marketing communication	1,134	80%	4%
Management assistant for dialog marketing	774	35%	7%
Management assistant for audiovisual media	180	92%	2%
Digital & Print Media Specialist	2,169	64%	6%
Bookseller	504	78%	1%
Advertisement, marketing, and media occupations	4,761	66%	5%

Source: own calculations based on Kroll, Schmidt and Uhly (2025).

Unfortunately, the official data does not contain detailed biographical information on the educational pathways, which means that it is not possible to determine how many apprentices may have started but not completed higher education. This limits the ability to estimate the prevalence of college dropout in the actual applicant pool of the FE. Drawing on general empirical findings, about 30% of Bachelor student's dropout and approx. 40% of them become apprentices afterwards (Neugebauer, Heublein and Daniel, 2019). Considering the 479,000 first semester students at German universities (of applied sciences), this translates in around 60,000 switchers to vocational education and training. Compared to the total of 480,000 new trainees in 2023 (Statistisches Bundesamt, 2024), this corresponds to about 12% of all new contracts. Under the assumption that these switchers have the same application rate and hiring probability than anybody else in the applicant pool regardless of the comparably higher educational attainment, 1 out of 8 applicants is a former student. In the presented FS, 50% of

all vignettes were college dropouts (25% intermediate and the other 25% possess Abitur only). Although, this distribution is needed to construct an orthogonal and balanced experimental design, it raises concerns about the ecological validity of the FS with respect to the assumed distribution of educational signals in the population.

In order to test the labour queue–derived expectation that the dropout penalty is more pronounced in fields where higher education is more prevalent, we replicate the regression models presented in Forster and Neugebauer (2024), but additionally allow the estimate of interest to vary by occupational field through an included interaction term. Results are displayed in Figure 1 and calculated as average marginal effects.



Source: own calculations based on the publicly available data from Forster & Neugebauer (2024), <https://osf.io/x2tcp>. The syntax file to reproduce these results can be found here: <https://osf.io/qac86>.

Figure 1. Effect heterogeneities for the 'dropout'-effect by experiment and occupational field.

On the one hand, the estimated penalty for higher education dropout against Abitur only applicants for electronic apprenticeship occupations, which is the field with the lowest expected proportion of applicants with Abitur (with or without dropout), is essentially zero in both experiments. The p-value displayed on the right-hand side of the graph ($p=0.883$) indicates that the estimates from both experimental studies do not differ significantly. Applicants for laboratory or administration occupations on the other hand, face a dropout malus in the FS

only. Both statistical tests confirm that the differences between the FE and FS estimates are statistically significant ($p=0.061$ for laboratory and $p=0.014$ for administration occupations). In both occupational fields the majority of hired apprentices hold upper secondary school-leaving certificates. While we do not want to overemphasize the difference between the two FS estimates, the direction is as expected (-0.12 vs -0.06), assuming a higher penalty in the occupational field with even higher proportions of highly qualified applicants. All three findings are in line with our hypothesis.

The only deviation from the expected estimate pattern is found in the media occupations, where no differences emerge either relative to the null hypothesis or between the experiments ($p=0.153$). However, even though this is in terms of respondents the second largest occupational field, the comparably large confidence interval of the estimate indicates a lot of variation in the ratings. Based on the official statistics this is also the most heterogeneous occupational field in terms of the school-leaving qualifications of the new apprentices (cf. Tab. 1). Unfortunately, we are unable to further decompose the actual sample at the occupation level and examine whether this result could be driven (in part) by disproportionately large sample proportions of occupations with a less qualified applicant pool (e.g. Management assistant for dialog marketing).

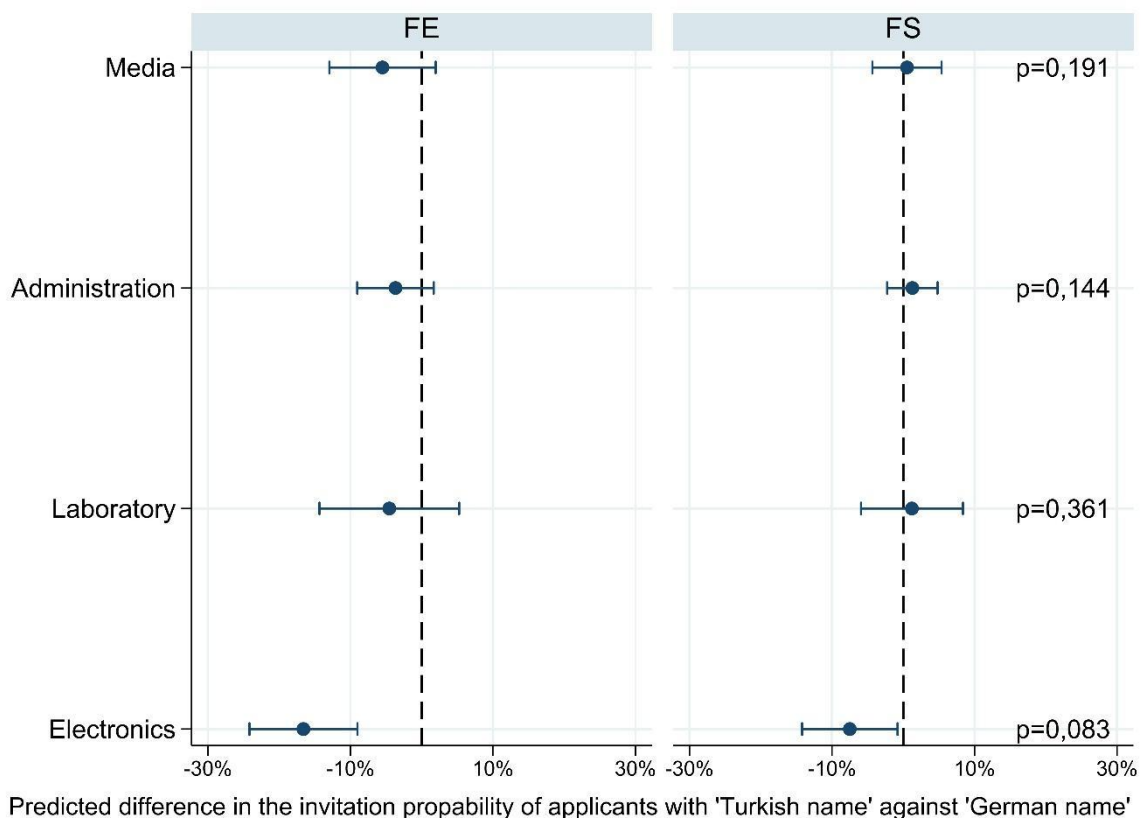
Additional analyses on effect heterogeneity for the ethnic background

In their validation study, Forster and Neugebauer (2024) also investigate the hiring value of a 'sensitive' factor. They argue from an empirical perspective that discrimination in hiring based on ethnicity has been observed in various fields and also in the context they studied, although it is socially inappropriate and illegal to consider an applicant's ethnic background when hiring employees (Forster and Neugebauer, 2024: 890). Therefore, the expectation is a more pronounced difference between the FE and FS estimates than with the non-sensitive educational signal and they indeed document a significant discrimination in the FE (-0.07) and basically the same hiring probability for applicants with German and Turkish names in the FS.

Although the focus of their validation study is on comparing the results between the two experiments, it is worth taking a theoretical look at the question of why employers discriminate on the basis of ethnic background. While Becker (1971) assumes that employers discriminate based on their preferences, proponents of statistical discrimination assume that employers have group-related productivity expectations and also modify these on the basis of their own experiences (Aigner and Cain, 1977; Arrow, 1973; Phelps, 1972). Ethnic discrimination in hiring can also occur due to prejudice arising from reduced contact between the employers and applicants from ethnic minorities (Pettigrew, 1998; Allport, 1954). There is empirical evidence suggesting different apprenticeship entry rates between Germans with and without

migration background based on their occupational interests (Beicht and Walden, 2015). This might on average result in occupation-specific differences in employer experience with applicants and former trainees from ethnic minorities.

However, we relax the implicit homogeneity assumption underlying Forster and Neugebauer's (2024) analyses and instead assume, that employers in the four occupational fields under study might vary in their contact and experience with applicants from different ethnic backgrounds and thus might have different stereotypical productivity-related expectations.



Source: own calculations based on the publicly available data from Forster & Neugebauer (2024), <https://osf.io/x2tcp>. The syntax file to reproduce these results can be found here: <https://osf.io/qac86>.

Figure 2. Effect heterogeneities for the 'discrimination'-effect by experiment and occupational field.

Figure 2 shows the results for this discrimination effect heterogeneity analyses. Although all field-specific estimates in the FE are negative indicating at least a tendency towards ethnic discrimination in all occupations, the only significant effect can be tested in electronic apprenticeships. The FS results mirror this pattern, likewise showing significant discrimination in electronic occupations, while the other three field-specific FS estimates are essentially zero. Wald tests (in the right-hand panel of Fig. 2) confirm that FE and FS estimates are statistically

comparable. This similarity is completely hidden in the main analyses, in which only the average effect is examined (Forster and Neugebauer, 2024: 896, Figure 3a).

As mentioned above, one possible explanation for the discrimination effect in electronic occupations might be different contact rates with apprentices of foreign origin in the past. Unfortunately, the official statistic nor other sources do provide reliable information on apprentice's ethnic background. The reported foreign origin (see Tab. 1) is calculated for apprentices without a German citizenship (Uhly, Schmidt and Kroll, 2024). If the trainee also has German citizenship, which should be the case for a majority of applicants with Turkish family roots, they are recorded as German. Additional insights might be generated by future research. All in all, the FS results do mimic the effect structure of the FE quite well when effect heterogeneities are considered.

Concluding remarks

The analyses presented in this replication study challenge the central conclusion drawn by Forster and Neugebauer (2024), namely that factorial surveys (FSs) are not suited for studying hiring behavior. Our findings point to the need for a more differentiated assessment, as a general conclusion may be premature. While we acknowledge the limitations inherent to FS designs, particularly regarding the ecological validity of applicant pools and the balance of experimental conditions, we argue that these limitations do not invalidate the method per se. In fact, these conditions can also be varied through adding different anchor vignettes to a constructed design and thereby experimentally vary the applicant pool.

Even if these conditions are not embedded in the design construction, our analyses show that when context-specific heterogeneities are taken into account, FSs can capture structurally meaningful patterns that align with real-world hiring behavior. In particular, we demonstrate that effect heterogeneities by occupational field reveal a more differentiated and theoretically plausible picture than average effects alone suggest. These findings highlight the importance of considering structural conditions when interpreting experimental data.

Ultimately, we do not recommend an uncritical endorsement of FSs, but caution against rejecting the method entirely based on average-level comparisons that may overlook underlying complexities. Future research should aim to improve the ecological validity of FS designs and incorporate context-sensitive modeling strategies that better reflect the diversity of real-life decision-making environments.

Declarations

Availability of data and materials

All analyses can be reproduced. The original dataset by Forster & Neugebauer (2024) can be found here: <https://osf.io/x2tcp>. The analysis syntax for this replication study is available here: <https://osf.io/qac86>.

Competing interests

The authors declare that they have no competing interests.

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Social impact & responsibility

This study revisits conclusions about the validity of factorial surveys in the context of hiring decisions, a domain with direct societal relevance, particularly regarding educational inequality and ethnic discrimination. Our reanalysis highlights that both educational signals and ethnic background effects are context-dependent and vary across occupational fields rather than reflecting uniform patterns. This has two important implications. First, findings on hiring discrimination—especially those related to ethnic origin—should not be interpreted as universally absent or present based on aggregate effects alone. Misinterpretations of such average results could either understate existing inequalities or overgeneralize discrimination patterns across contexts. Second, our results caution against drawing broad methodological conclusions (e.g., dismissing factorial surveys entirely) without accounting for effect heterogeneity and structural conditions.

At the same time, the study acknowledges limitations that are relevant for responsible interpretation. In particular, factorial surveys may rely on stylized and potentially unrepresentative applicant pools, which can affect ecological validity. Our own analyses demonstrate that such design features interact with real-world structures, such as occupational composition, and thereby shape estimated effects. Consequently, applying findings from factorial surveys to policy or practice requires careful consideration of context specificity.

Finally, because the study engages with sensitive attributes such as ethnic background, it is important to emphasize that observed differences should not be interpreted as reflecting inherent group characteristics but rather as outcomes of social processes, including employer perceptions and structural inequalities. By foregrounding heterogeneity and contextual variation, this study aims to contribute to a more nuanced and responsible understanding of hiring behavior and methodological inference in inequality research.

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